Concepts

Events

Nautilus is event-driven: every state change in the system is represented by an event object that flows through the `MessageBus` to strategy and actor handlers. This guide covers the event types, how they are dispatched, and how order fills produce position events.

Event categories

Category	Examples	Origin
Order	`OrderAccepted`, `OrderFilled`, `OrderCanceled`	`ExecutionEngine` (from venue)
Position	`PositionOpened`, `PositionChanged`	`ExecutionEngine` (from fills)
Account	`AccountState`	`ExecutionClient` / `Portfolio`
Time	`TimeEvent`	`Clock` (timers and alerts)

Handler dispatch

When an event reaches a strategy, the system calls handlers in a fixed priority order. The first matching handler runs, then the next level, so you can handle events at whatever granularity you need.

Order events

1. Specific handler (e.g. `on_order_filled`)
2. `on_order_event` (receives all order events)
3. `on_event` (receives everything)

Position events

1. Specific handler (e.g. `on_position_opened`)
2. `on_position_event` (receives all position events)
3. `on_event` (receives everything)

Time events

Timers and alerts produce `TimeEvent` objects. Pass a `callback` when calling `set_timer` or `set_time_alert` to direct events to your own method. If you omit the callback, the event is delivered to `on_event` instead.

Order events

Each order event corresponds to a state transition in the order state machine. The `ExecutionEngine` applies the event to the order, updates the `Cache`, and publishes it on the `MessageBus`. The table below shows the primary transitions; partially filled and triggered orders support additional transitions documented in the full order state flow.

Event	Primary transition	Handler
`OrderInitialized`	(created locally)	`on_order_initialized`
`OrderDenied`	Initialized -> Denied	`on_order_denied`
`OrderEmulated`	Initialized -> Emulated	`on_order_emulated`
`OrderReleased`	Emulated -> Released	`on_order_released`
`OrderSubmitted`	Initialized/Released -> Submitted	`on_order_submitted`
`OrderAccepted`	Submitted -> Accepted	`on_order_accepted`
`OrderRejected`	Submitted -> Rejected	`on_order_rejected`
`OrderTriggered`	Accepted -> Triggered	`on_order_triggered`
`OrderPendingUpdate`	Accepted -> PendingUpdate	`on_order_pending_update`

Event	Primary transition	Handler
`OrderPendingCancel`	Accepted -> PendingCancel	`on_order_pending_cancel`
`OrderUpdated`	PendingUpdate -> Accepted	`on_order_updated`
`OrderModifyRejected`	PendingUpdate -> Accepted	`on_order_modify_rejected`
`OrderCancelRejected`	PendingCancel -> Accepted	`on_order_cancel_rejected`
`OrderCanceled`	PendingCancel/Accepted -> Canceled	`on_order_canceled`
`OrderExpired`	Accepted -> Expired	`on_order_expired`
`OrderFilled`	Accepted -> Filled/PartiallyFilled	`on_order_filled`

Common order event fields

All order events share these fields:

Field	Description
`trader_id`	Trader instance identifier.
`strategy_id`	Strategy that submitted the order.
`instrument_id`	Instrument for the order.
`client_order_id`	Client-assigned order identifier.
`venue_order_id`	Venue-assigned order identifier.
`account_id`	Account the order belongs to.
`reconciliation`	Whether generated during reconciliation.
`event_id`	Unique event identifier.
`ts_event`	Timestamp when the event occurred.
`ts_init`	Timestamp when the event was created.

Individual events add type-specific fields (e.g. `OrderFilled` adds `last_qty`, `last_px`, `trade_id`, `commission`). See the API reference for the full field list per event type.

> Override `on_order_event` to handle all order events in one place. The specific handlers fire first, so you can mix both approaches.

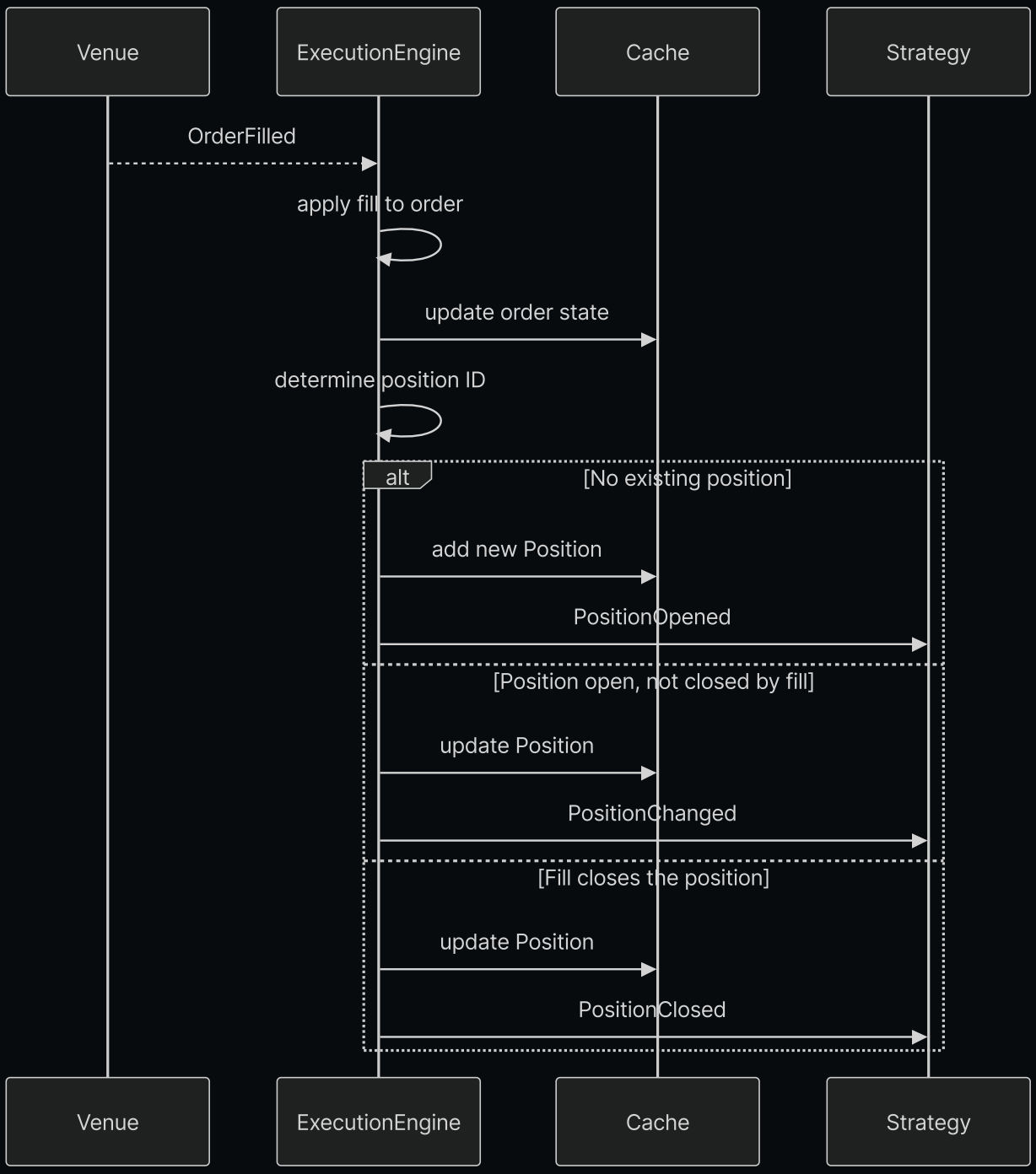
Position events

Position events are a direct consequence of fill events. The `ExecutionEngine` processes each `OrderFilled`, updates or creates a position, and emits the corresponding position event.

Event	When it fires	Handler
`PositionOpened`	First fill creates a new position.	`on_position_opened`
`PositionChanged`	Subsequent fill changes quantity or side.	`on_position_changed`
`PositionClosed`	Fill reduces quantity to zero.	`on_position_closed`

From fill to position: the causal chain

The following diagram shows how a single `OrderFilled` event produces a position event. This is the key link between order management and position tracking.



Step by step:

1. **Fill arrives.** The `ExecutionEngine` receives an `OrderFilled` event from the venue adapter.
2. **Order state updates.** The engine applies the fill to the order object and writes the updated order to the `Cache`.
3. **Position ID resolved.** The engine determines which position this fill belongs to, based on OMS type and strategy configuration.
4. **Position created or updated.** Three outcomes:
   - **No position exists** for this ID: the engine creates a `Position` from the fill, adds it to the `Cache`, and emits `PositionOpened`.

- **Position exists and remains open** after the fill: the engine applies the fill to the position, updates the `Cache`, and emits `PositionChanged`.
- **Position exists and closes** (quantity reaches zero): the engine applies the fill, updates the `Cache`, and emits `PositionClosed`.

5. **Flip case.** When a fill reverses the position (e.g. long 10 filled sell 15), the engine splits the fill into two parts: one that closes the original position (`PositionClosed`) and one that opens the new position (`PositionOpened`).

Position event fields

Field	Opened	Changed	Closed	Description
`trader_id`	✓	✓	✓	Trader instance identifier.
`strategy_id`	✓	✓	✓	Strategy that owns the position.
`instrument_id`	✓	✓	✓	Instrument for the position.
`position_id`	✓	✓	✓	Unique position identifier.
`account_id`	✓	✓	✓	Account the position belongs to.
`opening_order_id`	✓	✓	✓	Order that opened the position.
`closing_order_id`	-	-	✓	Order that closed the position.
`entry`	✓	✓	✓	Side of the opening fill.
`side`	✓	✓	✓	Current position side.
`signed_qty`	✓	✓	✓	Signed quantity (negative=short).
`quantity`	✓	✓	✓	Unsigned position quantity.
`peak_qty`	-	✓	✓	Largest quantity held.
`last_qty`	✓	✓	✓	Quantity of the last fill.
`last_px`	✓	✓	✓	Price of the last fill.
`currency`	✓	✓	✓	Settlement currency.

Field	Opened	Changed	Closed	Description
`avg_px_open`	✓	✓	✓	Average entry price.
`avg_px_close`	-	✓	✓	Average exit price.
`realized_return`	-	✓	✓	Realized return as a ratio.
`realized_pnl`	-	✓	✓	Realized profit and loss.
`unrealized_pnl`	-	✓	✓	Unrealized profit and loss.
`duration_ns`	-	-	✓	Time held in nanoseconds.
`ts_opened`	-	✓	✓	Timestamp when position opened.
`ts_closed`	-	-	✓	Timestamp when position closed.
`event_id`	✓	✓	✓	Unique event identifier.
`ts_event`	✓	✓	✓	Timestamp of the triggering fill.
`ts_init`	✓	✓	✓	Timestamp when event was created.

Tracing orders to positions

The `Cache` provides methods to navigate between orders and positions:

```
# From a position, find all orders that contributed fills
orders = self.cache.orders_for_position(position.id)

# From an order, find the position it belongs to
position = self.cache.position_for_order(order.client_order_id)

# The opening order is stored directly on the position
opening_order_id = position.opening_order_id
```

Account events

`AccountState` events represent balance and margin snapshots. They fire when:

- The venue reports an account update (via the execution client).

- The `Portfolio` recalculates account state after a position update (for margin accounts with `calculate_account_state` enabled).

Account state contains balances, margins, account type, and base currency. The `Portfolio` subscribes to these events internally to maintain exposure and balance tracking.

Event subscriptions

Beyond strategy handlers, actors can subscribe to specific event streams for instruments they do not trade. These subscriptions use the `MessageBus` directly and do not involve the `DataEngine`.

Method	Handler	Receives
`subscribe_order_fills()`	`on_order_filled()`	All fills for an instrument.
`subscribe_order_cancels()`	`on_order_canceled()`	All cancels for an instrument.

These are useful for monitoring actors that track execution quality or fill rates across strategies without participating in order management.

For details and examples, see Order fill subscriptions and Order cancel subscriptions.

Related guides

- Orders - Order types and state machine.
- Positions - Position lifecycle and PnL.
- Execution - Execution flow and risk checks.
- Strategies - Handler implementations in strategies.
- Architecture - Data and execution flow patterns.

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